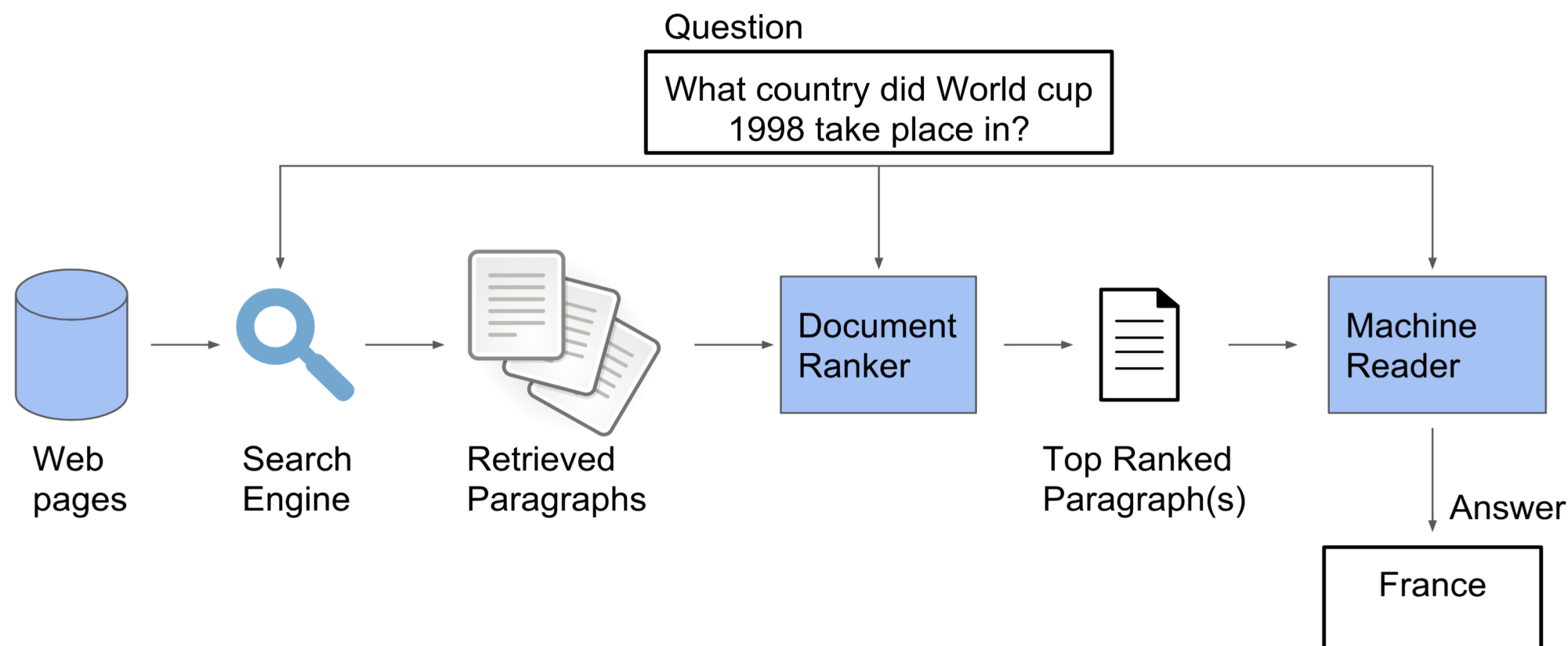
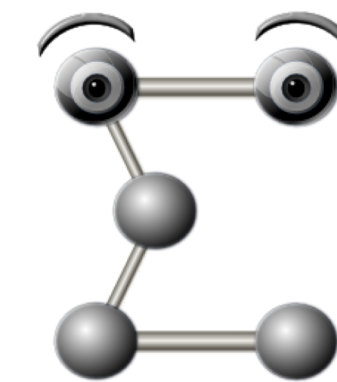




# Training a Ranking Function for Open-Domain Question Answering

Phu Mon Htut, Samuel R. Bowman, Kyunghyun Cho



## InferSent Ranker

- **Semantic similarity** based ranker
- We use *InferSent* [3], a sentence encoder trained on Natural Language Inference corpora, to encode the question and sentences in paragraph.
- Paragraphs are represented as sum of InferSent representations of sentences.

$$\mathbf{x}_{classifier} = \begin{bmatrix} q \\ p \\ q - p \\ q \odot p \end{bmatrix}$$
$$\mathbf{z} = \mathbf{W}^{(1)} \mathbf{x}_{classifier} + \mathbf{b}^{(1)}$$
$$score = \mathbf{W}^{(2)} ReLU(\mathbf{z}) + \mathbf{b}^{(2)}$$

## Machine Reader

- We use the Document Reader of DrQA [2], the open-domain QA system that combines search engine and neural-network based machine reader.
- Re-train DrQA on QUASAR-T dataset.

## Paragraph Selection

$$P(p_j^i, answer_j) = P(p_j^i) \times P(answer_j | p_j^i)$$

## Relation-Networks (RN) Ranker

- **Relevance-matching** based ranker
- We use adaptation of *Relation-Networks* [4], to measure the relevance between the words in questions and words in paragraph.

$$RN(q, p) = f_{\phi} \left( \sum_{i,j} g_{\theta}([E(q_i); E(p_j)]) \right)$$

## Training Criterion

- Margin ranking loss is used for both rankers:

$$\sum_{i=1}^k \max(0, 1 - f(q, p_{pos}) + f(q, p_{neg}^i))$$

## Recall of Rankers

|                   | Top-1       | Top-3       | Top-5       |
|-------------------|-------------|-------------|-------------|
| IR                | 19.7        | 36.3        | 44.3        |
| Ranker from $R^3$ | 40.3        | 51.3        | 54.5        |
| InferSent ranker  | 36.1        | 52.8        | 56.7        |
| RN ranker         | <b>51.4</b> | <b>68.2</b> | <b>70.3</b> |

## References

- [1] A deep relevance matching model for ad-hoc retrieval (Guo et al., 2017)  
[2] Reading Wikipedia to Answer Open-Domain Questions (Chen et al., 2017)  
[3] Supervised Learning of Universal Sentence Representations from Natural Language Inference Data (Conneau et al., 2017)  
[4] A simple neural network module for relational reasoning (Santoro et al., 2017)

## Ranking Measures

- **Semantic similarity**: Measures the overall similarity in meaning between a question and a document.
- **Relevance matching**: Measures the word or phrase level local interactions between pieces of texts in a question and a document. More important than semantic similarity for information retrieval [1].

## Dataset

- The QuesTion Answering by Search And Reading-Trivia (QUASAR-T),
- Addresses the combination of retrieval and machine reading.
- 43,013 open-domain questions based on trivia.
- Candidate passages in this dataset are collected from a Lucene based search engine, built upon ClueWeb09.

## Top passages returned by the two rankers for a question from test set

Question: Which country's name means "equator"? Answer: Ecuador

### InferSent ranker

**Ecuador** : "Equator" in Spanish , as the country lies on the Equator.

The equator crosses just north of **Ecuador's** capital, Quito, and the country gets its name from this hemispheric crossroads.

The country that comes closest to the equator without actually touching it is Peru.

The name of the country is derived from its position on the Equator.

### RN ranker

Salinas, is considered the best tourist beach resort in Ecuador's Pacific Coastline.. Quito, **Ecuador** Ecuador's capital and the country's second largest city.

Quito is the capital of **Ecuador** and of Pichincha, the country's most populous Andean province, is situated 116 miles from the Pacific coast at an altitude of 9,350 feet, just south of the equator.

The location of the Republic of **Ecuador** Ecuador, known officially as the Republic of Ecuador -LRB- which literally means "Republic of the equator" -RRB- , is a representative democratic republic

The name of the country is derived from its position on the Equator.

## Overall QA Performance

|                            | EM          | F1          |
|----------------------------|-------------|-------------|
| No ranker + DrQA           | 19.6        | 24.43       |
| InferSent + DrQA           | <u>31.2</u> | <u>37.6</u> |
| RN + DrQA                  | 26.0        | 30.7        |
| GA (Dhingra et al., 2017a) | 26.4        | 26.4        |
| BiDAF (Seo et al., 2016)   | 25.9        | 28.5        |
| $R^3$ (Wang et al., 2017)  | <b>35.3</b> | <b>41.7</b> |
| $SR^2$ (Wang et al., 2017) | 31.9        | 38.7        |

## Conclusion

- Interesting observation: Relation-Networks learn to predicts the answer on its own
- *Relevance matching based ranker* (Relation-Networks) provides **much better retrieval recall**
- But *semantic similarity based ranker* (InferSent ranker) has **better overall QA performance**
- High recall **doesn't always** translate to high overall QA performance